Breaking Bias. Building Futures.

ADDRESSING GENDER BIAS IN AI FOR EARLY EDUCATION

A Policy Brief



EQUITY.
EDUCATION.
EMPOWERMENT.
FOR EVERY GIRL.

POLICY RECOMMENDATIONS FOR INCLUSIVE, BIAS-FREE AI IN EDUCATION

Executive Summary

Educating our youth is critical to the future of our society. Young minds are impressionable, and Administrators implementing AI into school curriculum must screen for gender bias in education provided to advance future generations to a better world than the one we have today. Artificial intelligence learns from the data it is trained on, which may contain societal biases, requiring deliberate evaluation, good faith, and due diligence to mitigate them. Not doing so could further put young girls at risk, reinforcing gender stereotypes, reducing educational opportunities, widening the gender gap, and limiting equal participation in society.

Administrators cannot complete this task alone. The U.S. Department of Education has the responsibility to set key guidelines for early childhood educators to follow and developers of AI technologies in schools on how to create AI tools that reduce and mitigate biases. This should include requirements of transparent documentation from developers, regular auditing for bias in AI tools, and a commitment to dedicated research on how AI in early education supports or harms students. The primary audience for this policy brief is the U.S. Department of Education policymakers, more specifically, those within the U.S. Department of Education that are responsible for AI governance and educational technology policy.

Introduction

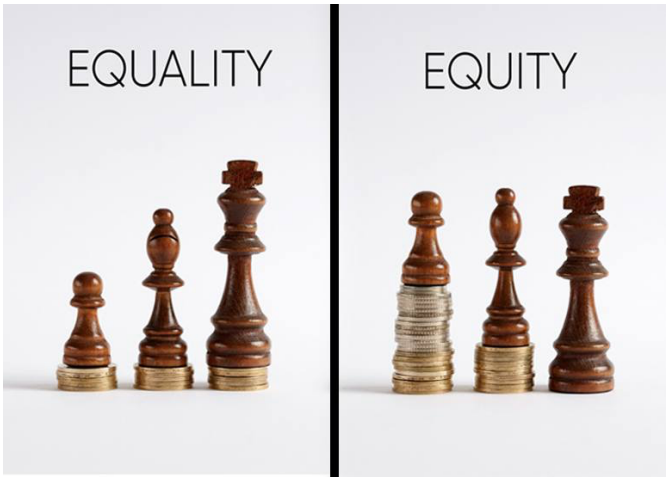
AI in education is becoming a standard practice as it rapidly enters classrooms across the nation. It can offer massive benefits in accessibility and personalization in learning, yet without significant safeguards against gender bias, it can unintentionally reinforce gender stereotypes or unequal learning opportunities. These systems are trained on historical data, which may reflect outdated societal biases, unintentionally harming students' learning paths.

The early years of a child's education are a critical period where they are learning not only about math, science, English, and the arts, but also about the world around them, their beliefs in how things work, and shaping their core values for the future. Their perception of the world is forming, and this is the time to ensure the next generation does not adopt the stereotypes and biases of the prior generations. AI is a critical function that can be utilized to either break these archaic and harmful ideas or further perpetuate them. If AI is permitted to further allow harmful biases to persist, it may unintentionally limit future opportunities for girls by reinforcing outdated assumptions about intellect and abilities solely based on gender.

This policy brief will outline the need for robust, cohesive, centralized policy guidelines that will identify and mitigate gender bias in AI technology within early education. Secondary to the primary audience, the U.S. Department of Education, additional policy actors include state and national education boards, school administrators, educators, and policymakers at the local, state, and national level. By establishing consistent, cohesive standards applicable to all early educational facilities, we can ensure young minds are receiving equal quality education everywhere. The call here is for transparent processes, accountability from AI technology developers, and further research to ensure AI technology is a positive addition for all students and not causing undue harm in the classroom.

Problem Statement

There is a lack of consistency in addressing gender bias within AI. As it continues to be implemented in education, it is imperative to address gender bias, particularly in early education, to prevent the continuation of harmful stereotypes of young girls and women. There are piecemeal efforts at varying degrees to address this bias, but that fragmentation can lead to confusion or lack of universal adoption, which will not prevent the gender gap from being reduced and may widen it. **"**Female students frequently face unequal access to education resources and engagement in educational interventions, including unequal opportunities and gender-biased social barriers" *(Kim & Kwon, 2024)*. This unequal access at a



young age compounds over time, potentially contributing to long-term effects, including lack of career pathways, earnings, and educational opportunities. Gender stereotypes constitute a serious obstacle to equality between women and men, contributing to gender segregation in education and employment, further widening the gender gap in the digital sector and preventing women's full participation, and these stereotypes are endorsed by, and influence the interests of, children as young as the age of six, specifically girls" (European Parliament, 2021). With no concrete policy in place for administrators and educators to follow, the AI being developed and deployed lacks meaningful guardrails and mitigations for bias in the classroom. The AI being used in one school could have vastly different effects on children than the AI in another, with no central source of truth.

Summary of current Policy and knowledge or research related to the problem

In researching active current policy regarding gender discrimination in early education and AI adoption, it has been found that the United States currently lacks a comprehensive federal framework governing gender bias testing in AI educational technologies. The only state resolution regarding bias that has been located is Illinois, which "urges the Illinois State Board of Education to develop statewide guidelines for the ethical and responsible use of artificial intelligence in education, including monitoring for bias, to ensure students can thoughtfully and ethically engage with emerging technologies" (Illinois General Assembly, 2026).

Policy projects targeting the reduction of gender stereotypes are not universal (Brussino & McBrien, 2022)**;** they highlight that these initiatives are carried out at different levels in

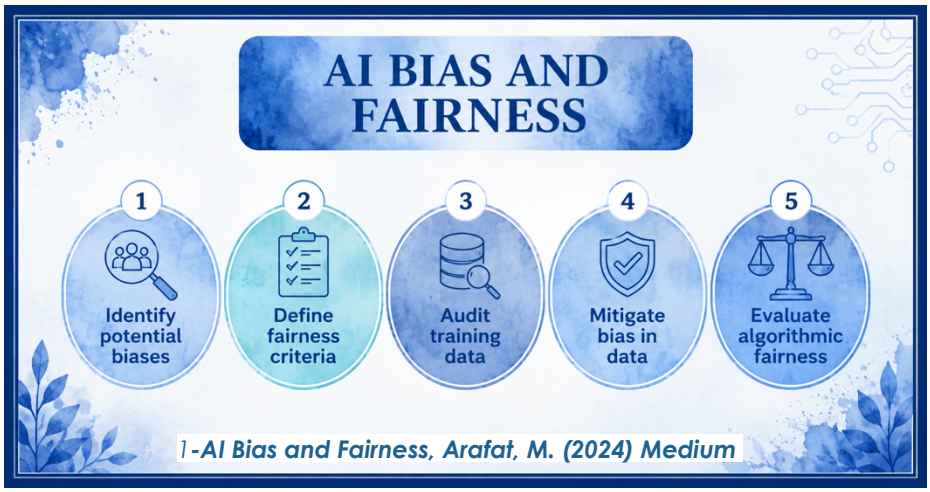
partnership with different educational stakeholders, which can result in fragmentation of results and a delay in progress.

Research shows that "there is increasing recognition that identifying and preventing damaging gender attitudes and stereotypes in childhood, before they become entrenched, is key to enacting broad social change" (King et al., 2021). This makes it imperative to address gender bias at an early age, and consistently through policy reform.

Currently, the only protections against discrimination in the United States are from 1972. Much has evolved and changed since then, but there have not been updates to protections from gender bias, and none in education to specifically address AI at the national level for early education. Further, Title IX only refers to educational programs that receive federal financial assistance, leaving young students open to discrimination if they do not receive federal funding(U.S. Department of Education, n.d).

Policy Options to Consider (Policy Options Under Consideration)

Some Policy options to consider include Placing requirements on the corporations that design AI systems to adopt UNESCO's policy measure proposal regarding "inclusion, equity, and linguistic and cultural diversity: Develop criteria for the validation of GenAI systems to ensure that there is no gender bias, discrimination against marginalized groups, or hate speech embedded in data or algorithms" (Miao & Holmes, 2023). By adopting UNESCO's policy measure proposal, educators and administrators will be empowered to take action and find support in regulating this fast-evolving technology to provide their students with the best tools available while combating gender bias in the critical learning years shaping young minds. Another option would



1-AI Bias and Fairness, Arafat, M. (2024) Medium

be to Align with the European Parliament's resolution and create a federally mandated policy to "propose initiatives, including research, to understand and address the root causes of the digital gender gap, collect and utilize existing sex-disaggregated data to promote more research into the interaction of the different factors which impede women's and girls' digital inclusion; stresses that harmonized data collection facilities the comparison and sharing of data and examples of best practices by Member States" (European Parliament, 2021). The EU has been leading the way with digital protection, from the GDPR to the EU AI Act, ensuring robust involvement of key stakeholders and keeping the protection of people at the forefront, showing trust in adopting their resolutions to fit the needs of the Department of Education. Aligning with the EU's resolution and creating a federally mandated policy would ensure continuity for schools across the nation, lessening the risk of bias mitigation efforts being applied unequally, causing

our young minds to receive varying qualities of education and exposure to unnecessary gender bias. Finally, another option to consider would be to require state-level AI governance and develop "guidelines for the ethical and responsible use of artificial intelligence in education, including monitoring for bias, to ensure students can thoughtfully and ethically engage with emerging technologies" (Illinois General Assembly, 2026). States could develop, mandate, and implement AI governance policies faster than at the federal level, but this could increase the risk of uneven or unequal exposure to gender bias depending on the state's political views and how seriously this real risk is taken.

Recommendations

1. The U.S. Department of Education should require all AI-enabled instructional technologies in educational institutions that integrate with early education to implement independent gender bias testing, periodic bias audits, teacher oversight, public reporting with transparent documentation of AI usage, and regular recertification with AI tools that include equal access to educational materials and a dedicated research team to find where bias exists to reduce and eliminate it.
2. School administrators in early education facilities should act and incorporate already developed guidelines, such as Illinois' State Resolution on Artificial Intelligence, to start with some regulations while waiting for finalized guidelines from the Department of Education.
3. Early Education Educators should implement regular checking and re-checking of AI tools for bias, with audits being sent to Administrators for review and contact with developers for action plans with deadlines to correct issues found.
4. The Department of Education should petition for research funding towards the study of gender bias in Early education and AI and how to reduce/eliminate it

Summary and Conclusions

Early education is a critical time for youth to absorb knowledge in the classroom about intentional topics and unintentional ones, like the world around them. Artificial Intelligence has amazing properties to propel the next generation ahead, supporting learning in ways unthought of previously, adding inclusion and personalized learning options never before available. Yet without concentrated effort to implement policy reducing bias, it could create untold harms and further reinforce outdated and unwarranted gender stereotypes. Concerted efforts to eliminate bias will enable young girls to materially participate and shrink the gender gap. This time of rapid technological advancement is a time when positive change can be made, and AI can be a catalyst for young girls to grow into a world that does not place her consistently behind or stereotyped. A move towards national policy standards now can prevent further widening of the gender gap, marking AI as a historical landmark for women's rights in a positive direction, rather than reinforcing historical gender disparities for future generations. Studies have shown "that stereotypes and confidence gaps, rather than ability, are primary factors shaping gendered participation" (Kim & Kwon, 2025).

Currently, at the national level, there is no comprehensive universal policy to hold AI educational developers accountable or audit standards. Establishing policies at this level will provide a universal and consistent experience for students and enable developers of AI systems to follow clear guidelines in their creation and maintenance of educational materials.

The decisions we make today will hold weight in the next generation to come. Proactive policies now can provide a brighter future for our youth and prevent further generations from experiencing discrimination. AI can become a force for good if we take action now. Every child deserves an equal opportunity to learn and succeed, regardless of gender.

References

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Appendix A: Revise and Resubmit Memo

Dear Reviewers,

Thank you for the feedback on my draft! I appreciated the helpful information to strengthen this important topic. Here you will find a summary of the changes I've made, and what I've chosen not to change and why:

- ***First, all the reviewers noted a few of my in-text citations that were removed.*** *I had desired to switch to footnotes and realized that the newer version was not saved properly, resulting in submitting a draft with half APA and half missing citations- I updated this on pages 3, 4, and 5, and stuck with APA to be safe. One additional note here: reviewers mentioned not seeing Arafat in the body of the paper; however, Arafat is cited on the image it was utilized from, and I've listed this in the bibliography as a credit to the image. I've since removed Arafat from the bibliography but left the credit on the photo on page 4 to ensure author credit.*
- ***Two reviewers mentioned tightening up the recommendations section****. One highlighted scope creep, and another highlighted different actors. I tightened up this section on page 5 to make it clear that each recommendation tied back to early education, and how each was a realistic option for administrators or educators to act now if bureaucratic agencies were slow to take action on this critical issue.*
- ***One reviewer suggested revisiting the Policy Options Sections to add additional strength****. I revisited the Policy Options section on page 4 and detailed out why the options would be beneficial and what potential drawbacks, if any, could result from taking that measure.*
- ***There were two Suggestions I did not adopt****. The first I found confusing, as it stated, "the executive summary is not visually set apart from the rest of the brief" and suggested a shaded background or border around it. I did not adopt this suggestion, because my executive summary did have this, and in the color theme of the document. A lightish blue rounded edge box around the Executive Summary was very clearly within the draft and is still within this final submission. Second, a reviewer commented requesting the conversation itself from the AI chat; however, the conversation is output from the images shown in the text, and the AI memo states where to find those images. I appreciate all of the feedback provided to help make this final output stronger.*

Appendix B: Final Writing Process Memo & AI Usage Memo

In writing this policy brief, I continued focusing on the topic I started researching during the Annotated Bibliography and Policy Brief outline assignment we had a few weeks ago, considering feedback from that assignment to refine further and structure the policy brief. To start, I ported over my Policy Brief outline and references to review the work I had already completed and began formatting the bullet points into a more formalized and digestible format. Reviewing the references in more detail, I not only extracted key details, but learned quite a bit along the way. I did not know that Title IX only refers to educational programs that receive federal financial assistance, and it is from 1972 to address gender bias. This further fueled the desire to learn more, find out what is being done, and provide succinct information within this brief, ensuring those who do read it are inspired not just to learn more, but to act.

I utilized AI in this Policy Brief to create graphics and images to catch the attention of the reader, including the title page, by inputting my draft and asking the AI to output an image. I did attempt to also adjust the equality/equity image to match the color theme of the brief, but it failed spectacularly, showing up in pink, purple, and green, placing the platforms at the wrong

heights, and it even added the young girls, but it could not get the image correct. I had to forgo that option. I used the same prompt as the one for the other in-text image: "Update this image in the style of the title page:" and pasted the image in.

Prompt:
Here is a draft version of my policy brief. I've chosen the color blue as my border and heading colors, as a nod to breaking gender stereotypes (the stereotype is that girls are pink and boys are blue)

Output:
(See cover page)

Prompt:
Update this image in the color profile and style of the title page:

Output:
(Updated image of AI bias and fairness)